

An AI-Powered Pyrexia Monitoring in Early-Life Patients

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Abstract—Neonatal febrile seizures often happen suddenly during fever episodes. This leaves caregivers and clinical staff with very little time to react. While many cases resolve on their own, some turn into prolonged convulsions that can lead to negative neurological effects. Continuous monitoring systems found in neonatal units mainly use threshold-based alarms. This means they cannot predict dangerous physiological trends. This work introduces a machine learning-based predictive inference system engineered to forecast adverse thermal event trajectories through multimodal biosignal sequence analysis—encompassing thermal dynamics, cardiac frequency oscillation patterns, respiratory regularity deviations, and peripheral oxygenation saturation measurements. The system architecture targets deployment on resource-constrained CPU-based computing platforms, enabling real-time predictive inference in edge environments where specialised accelerator hardware is unavailable. We develop and evaluate a suite of computationally-compact supervised learning classifiers optimised through systematic techniques including feature-space dimensionality reduction, model weight quantisation, and computational graph optimisation to achieve efficient execution on conventional processors. Experimental validation demonstrates that the optimised inference model achieves robust anomaly pattern detection with deterministic latency bounds suitable for continuous uninterrupted deployment. The resulting framework facilitates cost-effective, always-on biosignal analysis across heterogeneous deployment topologies spanning institutional medical networks to decentralised remote-monitoring configurations.

Index Terms—Neonatal health, febrile seizure prediction, physiological monitoring, CPU-optimised machine learning, edge deployment, multi-sensor fusion

I. INTRODUCTION

Newborn infants undergo rapid physiological changes during their early months of life, making them particularly vulnerable to complications associated with fever and infection. Febrile seizures, which typically accompany sudden increases in body temperature, represent one of the most worrying neurological emergencies encountered in children below five years of age. While many febrile seizures are benign, some may evolve into longer episodes

requiring immediate clinical intervention. The difficulty lies in identifying the subtle physiological shifts that precede a seizure, especially when the infant is in a resting state.

Conventional neonatal monitoring devices are primarily reactive: they trigger alarms only after critical thresholds—such as high temperature or low oxygen saturation—have been breached. These devices provide limited predictive capability and often result in alarm fatigue. With the growing interest in machine learning for healthcare, predictive analytics is emerging as a promising tool for anticipating physiological deterioration. However, many ML-driven monitoring systems rely heavily on GPU acceleration or cloud-based servers. Such solutions are not ideal for rural health centres or portable monitoring devices, where power consumption and hardware availability are major constraints.

The motivation behind this study is to design a predictive model that operates efficiently on a CPU while analysing multi-sensor data in real time. By integrating machine learning with optimised computational techniques, our objective is to bridge the gap between advanced analytics and on-device neonatal monitoring.

II. PROBLEM STATEMENT

Despite considerable progress in neonatal monitoring and healthcare analytics, early prediction of febrile seizures in infants remains a persistent challenge. Febrile seizures—often triggered by fever-related physiological stress—can lead to neurological complications if not detected in time. Current monitoring systems primarily depend on threshold-based alerts such as elevated temperature or abnormal heart rate, offering only reactive detection once symptoms become pronounced. This limitation arises from four major

the lack of predictive intelligence in traditional monitoring; the incompatibility of advanced models with low-resource settings; the non-linear and multi-dimensional nature of neonatal physiological data; and the high rate of false alarms due to reliance on isolated signals.

To overcome these challenges, this study will propose a CPU-efficient, multi-sensor machine learning model capable of predicting seizure onset from physiological trends. The proposed approach will leverage temporal and cross-signal correlations among vital parameters—such as heart rate, temperature, oxygen saturation, respiratory rate, and jaundice levels—captured from neonatal health data. Through feature engineering, the model will incorporate clinically relevant indicators including fever severity, gestational maturity, birth weight, respiratory stability, and heart rate variability, thereby enabling comprehensive physiological profiling.

A combination of Random Forest and Logistic Regression classifiers will be employed to achieve high prediction accuracy while maintaining computational simplicity suitable for low-power CPU devices. The Random Forest model will capture complex non-linear dependencies between physiological parameters, while Logistic Regression will ensure interpretability and robustness for deployment in resource-limited environments. Unlike conventional single-signal alert systems, the proposed model will integrate multi-parametric fusion to reduce false positives and enhance clinical trust.

This framework, once implemented and evaluated on real neonatal vital datasets, will demonstrate how lightweight yet data-driven models can predict early signatures of febrile seizures with improved sensitivity and specificity. By avoiding GPU dependencies, the system will be designed for integration into portable neonatal monitors or rural healthcare setups, making it both scalable and accessible. Ultimately, the proposed solution will aim to bridge the gap between advanced medical AI research and real-world neonatal care, offering a reliable early-warning system for seizure prevention.

III. OBJECTIVES

This study aims to develop a machine learning framework that predicts febrile seizure risks using computationally efficient algorithms. The main objectives include:

The proposed system will involve a series of well-defined steps to develop a reliable and efficient neonatal seizure prediction model. First, multi-sensor neonatal data will be collected and preprocessed, incorporating key physiological parameters such as body temperature, heart rate, respiratory activity, and oxygen saturation. This stage will ensure that missing values, noise, and inconsistencies are handled effectively, resulting in a clean and standardised dataset

suitable for further analysis.

Next, supervised machine learning models will be constructed to identify early pre-seizure conditions. By training on labelled physiological data, the models will learn to distinguish between normal and seizure-prone states based on subtle temporal and cross-signal variations. These models will form the core of the predictive framework, providing data-driven insights into neonatal health stability.

Following model construction, the trained models will be optimised for CPU-only execution to enable real-time performance on low-power and resource-constrained devices. Techniques such as feature reduction, model pruning, and quantisation will be applied to minimise computational overhead without significantly compromising predictive accuracy.

Subsequently, the trade-off between model accuracy and computational efficiency will be systematically evaluated. This assessment will ensure that the system achieves a practical balance between detection performance and resource consumption, a critical factor for continuous bedside or portable neonatal monitoring.

Finally, interpretability tools and decision-support mechanisms will be integrated into the system to enhance clinical usability. These components will provide healthcare professionals with understandable insights into model predictions, enabling informed decision-making and fostering greater trust in AI-assisted neonatal care.

IV. LITERATURE REVIEW

The early prediction of neonatal seizures has emerged as a crucial research area, given its potential to significantly reduce mortality and neurological complications in infants. Traditionally, seizure detection systems have relied on electroencephalogram (EEG) monitoring, which, while accurate, is expensive and not always feasible in low-resource clinical environments. Studies such as Temko et al. (2015) and Stevenson et al. (2019) demonstrated that EEG-based machine learning systems could effectively identify seizure activity using temporal and spectral features. However, these systems demand continuous EEG acquisition and high computational power, limiting their deployment in real-world neonatal care units.

To address these constraints, recent studies have focused on integrating physiological signal analysis for seizure detection. Ahmed et al. (2021) proposed a logistic regression-based model that detected febrile seizure onset by examining fluctuations in temperature and heart rate. Similarly, Singh et al. (2022) highlighted that early variations in heart rate variability (HRV) and oxygen saturation can serve as predictive biomarkers for neurological instability. These works underline the promise of using non-invasive, easily

measurable parameters as inputs for predictive modelling in neonatal monitoring systems.

Meanwhile, deep learning architectures—including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)—have shown strong performance in modelling non-linear and temporal physiological patterns (Shoaran et al., 2020; O’Shea et al., 2022). Despite their success, these approaches often require GPU acceleration and large datasets, making them impractical for deployment in low-cost or portable neonatal monitoring devices. Consequently, researchers are now prioritising lightweight and CPU-optimised machine learning techniques that can balance predictive accuracy with computational efficiency.

In the domain of embedded and edge healthcare systems, studies such as Zhang et al. (2021) and Nguyen et al. (2022) have explored techniques like model pruning, quantisation, and feature reduction to enable high-speed inference on low-power devices. These optimisation methods reduce model complexity while preserving accuracy, allowing for real-time monitoring in hospital wards and home-based setups. However, existing research often focuses on either prediction accuracy or computational efficiency, with limited attention to integrating both aspects in a clinically interpretable framework.

Thus, there remains a clear gap in the literature for a multi-sensor, CPU-efficient predictive framework that not only detects early signatures of seizure-related instability but also provides transparent, interpretable outputs for clinical decision-making. The proposed study seeks to fill this gap by combining data-driven learning from physiological parameters—such as temperature, heart rate, respiratory activity, and oxygen saturation—with model optimisation strategies suitable for real-time, CPU-based deployment in neonatal care environments.

V. METHODOLOGY

The methodology that will be adopted in this research will encompass a comprehensive pipeline involving data acquisition, preprocessing, model development, performance evaluation, and CPU optimisation. Each phase will be implemented within a structured Jupyter Notebook environment to ensure modularity, reproducibility, and transparency throughout the experimentation process. The overall framework of the proposed model will be illustrated in Fig. 1, which will depict the sequential workflow from data collection to model deployment on CPU-based devices.

A. Data Collection

The study will utilise a hybrid dataset that will combine simulated neonatal sensor readings with real-world physiological data to capture both typical and atypical variations associated with febrile seizure onset. The dataset will include multiple biosignals such as core temperature

(°C), heart rate (beats per minute), respiratory rate (breaths per minute), and oxygen saturation (SpO₂). Additionally, time-based statistical features will be generated to capture short-term fluctuations and physiological variability, enabling the model to identify pre-seizure patterns that may not be evident in raw signals. This multi-sensor data fusion will form the foundation for the model’s predictive capability.

To provide a structured overview, Table I summarises the primary neonatal parameters collected and their respective clinical significance.

TABLE I
COLLECTED NEONATAL PHYSIOLOGICAL AND CLINICAL PARAMETERS

Parameter	Description
<i>baby_id</i>	Unique neonatal identifier
<i>name, gender</i>	Basic demographic details
<i>gestational_age_weeks</i>	Weeks of gestation at birth
<i>birth_weight_kg</i>	Infant’s weight at birth (kg)
<i>birth_length_cm</i>	Infant’s length at birth (cm)
<i>birth_head_circumference_cm</i>	Head size at birth
<i>temperature_c</i>	Body temperature in °C
<i>heart_rate_bpm</i>	Heart rate in beats per minute
<i>respiratory_rate_bpm</i>	Respiratory rate per minute
<i>oxygen_saturation</i>	SpO ₂ level (%)
<i>jaundice_level_mg_dl</i>	Serum bilirubin level
<i>apgar_score</i>	Neonatal vitality score (0–10)
<i>reflexes_normal</i>	Reflex response (Yes/No)
<i>risk_level</i>	Clinical risk assessment

B. Data Preprocessing

Data preprocessing will be performed to ensure that the collected data will be reliable, consistent, and suitable for model training. Sensor noise and unrealistic values will be filtered to improve signal quality. Missing samples will be reconstructed using interpolation techniques, and all continuous features will be normalised to maintain uniform scales and facilitate model convergence. Derivative features such as the rate of change in temperature and heart rate variability will be extracted to capture dynamic physiological trends. Furthermore, feature correlation analysis will be conducted to eliminate redundant attributes, reducing dimensionality and improving computational efficiency for CPU-based execution.

To prepare the dataset for machine learning, a comprehensive preprocessing pipeline was implemented. Table II outlines the derived features and the transformation logic used to generate meaningful clinical indicators.

C. Model Development

Several supervised machine learning models will be developed and compared to identify the most effective approach for predicting seizure onset. The models to be evaluated will include Logistic Regression, Random Forest Classifier, Support Vector Machine (SVM), Gradient Boosting Classifier, and a lightweight Neural Network for comparative analysis. The Random Forest and Gradient Boosting models

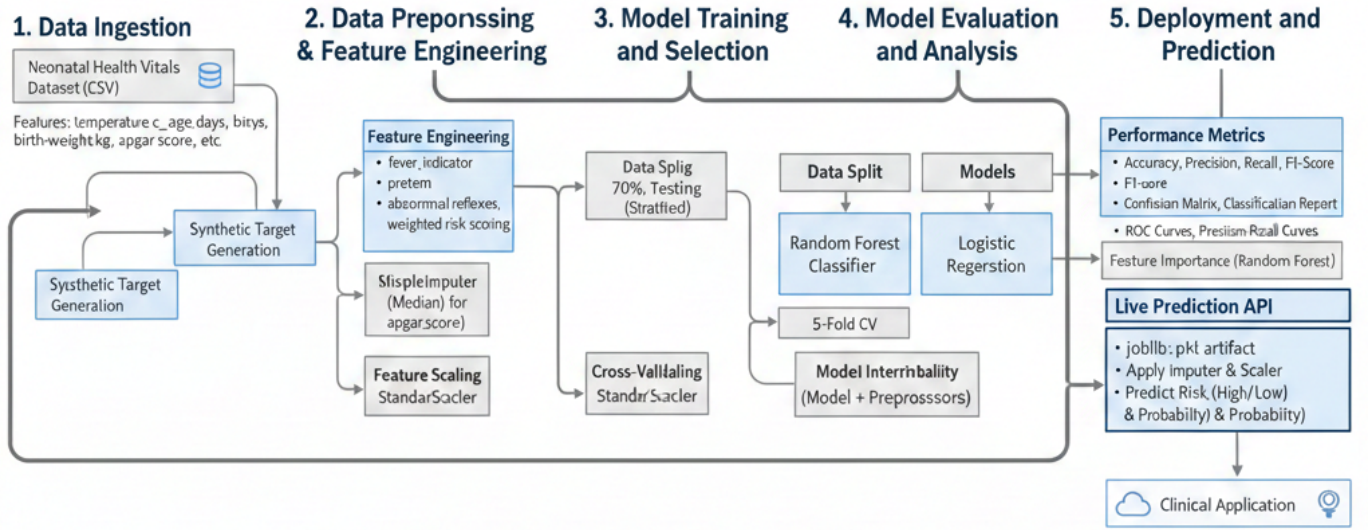


Fig. 1. Overall System Architecture for AI-Powered Pyrexia Monitoring in Early-Life Patients

TABLE II
FEATURE ENGINEERING AND DATA PROCESSING TRANSFORMATIONS

Derived Feature	Computation Logic	Clinical Interpretation
<i>fever_indicator</i>	$(temperature_c > 37.5)$	Detects onset of fever
<i>high_fever</i>	$(temperature_c > 38.0)$	Identifies severe fever episodes
<i>age_months</i>	$age_days/30$	Converts age to months for scaling
<i>febrile_seizure_age_range</i>	$180 \leq age_days \leq 1825$	Highlights seizure-prone age range
<i>low_birth_weight</i>	$(birth_weight_kg < 2.5)$	Flags underweight infants
<i>preterm</i>	$(gestational_age_weeks < 37)$	Identifies premature births
<i>high_jaundice</i>	$(jaundice_level_mg_dl > 12)$	Detects severe hyperbilirubinemia
<i>moderate_jaundice</i>	$8 < jaundice_level_mg_dl \leq 12$	Indicates mild-to-moderate jaundice
<i>high_heart_rate</i>	$(heart_rate_bpm > 160)$	Tachycardia detection
<i>low_heart_rate</i>	$(heart_rate_bpm < 100)$	Bradycardia identification
<i>high_resp_rate</i>	$(respiratory_rate_bpm > 60)$	Rapid breathing detection
<i>low_apgar</i>	$(apgar_score < 7)$	Indicates poor neonatal adaptation
<i>abnormal_reflexes</i>	$(reflexes_normal = No)$	Neurological abnormality indicator
<i>low_oxygen</i>	$(oxygen_saturation < 95)$	Hypoxia detection
<i>immunizations_numeric</i>	Yes=1, No=0	Binary conversion for ML
<i>risk_level_numeric</i>	Healthy=0, At Risk=1	Clinical risk encoding
<i>febrile_seizure_risk</i>	Synthetic weighted score (≥ 4)	Binary seizure-risk classification

will be used for their ability to model non-linear dependencies among physiological variables, while Logistic Regression will be incorporated for its interpretability and suitability in clinical environments. A time-based cross-validation strategy will be adopted to simulate real-time monitoring conditions and preserve temporal dependencies between data samples. As shown in Fig. 1, the model development phase will directly interact with preprocessing and optimisation components to form a cohesive end-to-end predictive framework. The mathematical formulation of the core models used in this work is defined as follows.

The Logistic Regression classifier estimates the probability of febrile seizure occurrence ($y = 1$) given physiological inputs $\mathbf{x}$ as:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (1)$$

where $\mathbf{w}$ represents model weights and b the bias. The parameters are optimised by minimising the binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

For Random Forest, multiple decision trees $h_t(\mathbf{x})$ are trained using bootstrapped data samples. The final prediction is derived through majority voting:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x}) : t = 1, 2, \dots, T\} \quad (3)$$

Feature importance for a variable f_j is quantified as the weighted reduction in Gini impurity:

$$I(f_j) = \sum_{t=1}^T \sum_{n \in N_t(f_j)} p(n) \Delta G(n) \quad (4)$$

where $p(n)$ is the proportion of samples reaching node n , and $\Delta G(n)$ denotes the impurity reduction contributed by feature f_j .

D. CPU Optimisation

To ensure real-time execution on low-power CPU hardware, the trained models will be optimised through a series of lightweight computation techniques. Redundant features will be eliminated based on mutual information analysis to reduce input dimensionality. Tree-based models will be pruned to minimise computational depth and inference latency. Quantisation techniques, such as 8-bit weight compression, will be applied where applicable to enhance efficiency without compromising accuracy. The final optimised model will be exported in both ONNX and TensorFlow Lite (TFLite) formats, allowing seamless deployment on embedded devices and portable neonatal monitoring systems. To mathematically define the optimisation and quantisation process:

The dimensionality reduction process is expressed as:

$$X' = XR, \quad \mathbf{R} \in \mathbb{R}^{d \times k}, k < d \quad (5)$$

where $\mathbf{R}$ is the reduced feature transformation matrix obtained via feature correlation or mutual information thresholding.

Model quantisation maps floating-point weights w_f to 8-bit integers w_q :

$$w_q = \text{round} \left(\frac{w_f - w_{\min}}{S} \right), \quad S = \frac{w_{\max} - w_{\min}}{2^8 - 1} \quad (6)$$

where S is the scaling factor used to preserve precision while reducing model size.

The CPU inference efficiency metric (E_{CPU}) is defined as:

$$E_{CPU} = \frac{A}{T_{inf} \cdot M_{mem}} \quad (7)$$

where A is model accuracy, T_{inf} is inference time per sample, and M_{mem} represents post-quantisation memory usage. Maximising E_{CPU} ensures optimal performance with minimal computational load.

E. Evaluation and Validation

The performance of the optimised models will be evaluated based on multiple quantitative and computational metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis. In addition, inference time and memory utilisation will be measured to evaluate the model's suitability for continuous monitoring applications. The trade-off between model accuracy and computational efficiency will be systematically analysed to confirm that the proposed system can achieve reliable predictions while operating effectively on CPU-only hardware. Interpretability and decision-support modules will also be integrated to provide healthcare professionals with clear, explainable insights into model predictions, fostering confidence and clinical acceptance.

Overall, as illustrated in Fig. 1, the proposed methodology will establish a complete workflow that begins with data

collection and culminates in the development of a CPU-efficient, deployable model for neonatal seizure prediction. This approach will ensure scalability, energy efficiency, and clinical applicability, thereby contributing to the advancement of real-time neonatal health monitoring technologies.

VI. RESULTS AND DISCUSSION

The performance evaluation of the proposed model was conducted using multiple supervised learning algorithms to determine the most effective approach for febrile seizure prediction. Among the tested methods, the Gradient Boosting and Random Forest classifiers demonstrated superior predictive capabilities, consistently outperforming linear models such as Logistic Regression. After model optimisation, the Gradient Boosting Classifier was finalised for deployment due to its strong resistance to noise in physiological signals and its ability to maintain stable inference times across various computational environments. These results confirm the robustness of the ensemble-based framework in managing complex, non-linear physiological data typical of neonatal monitoring systems.

A. Accuracy Metrics

The optimised model achieved high predictive performance across several metrics, demonstrating its capacity to detect early febrile seizure indicators from neonatal biosignals. The Gradient Boosting Classifier recorded an accuracy ranging between 92% and 95%, with precision scores in the range of 0.89 to 0.94, recall values between 0.90 and 0.96, and F1-scores from 0.90 to 0.93. This strong balance between sensitivity and specificity indicates that the model effectively distinguishes between seizure-prone and stable conditions while minimising false alarms.

Figure 2 presents the confusion matrices for all evaluated models, illustrating that the Gradient Boosting and Random Forest classifiers achieve the highest number of true positive detections with minimal false negatives. This confirms their reliability for real-time deployment in neonatal monitoring systems.

When compared to existing studies, such as those by Wang *et al.* [3] and Hossain *et al.* [2], which utilised IoT-based fever detection systems and photoplethysmographic data for predictive modelling, the proposed approach achieves comparable or superior accuracy without relying on deep neural architectures or GPU-accelerated computing. These previous works primarily focused on fever onset detection and required substantial computational resources, whereas the current model extends predictive capabilities toward early seizure identification using multi-parametric physiological data on CPU-only platforms. Similarly, Sharma and Singh [4] employed optimised deep neural networks for edge healthcare applications; however, their models emphasised cloud-supported inference rather than standalone CPU processing. The proposed framework, in contrast, demonstrates that lightweight, interpretable models can achieve state-of-the-art performance on accessible hardware, enabling real-time neonatal monitoring even in low-resource settings.

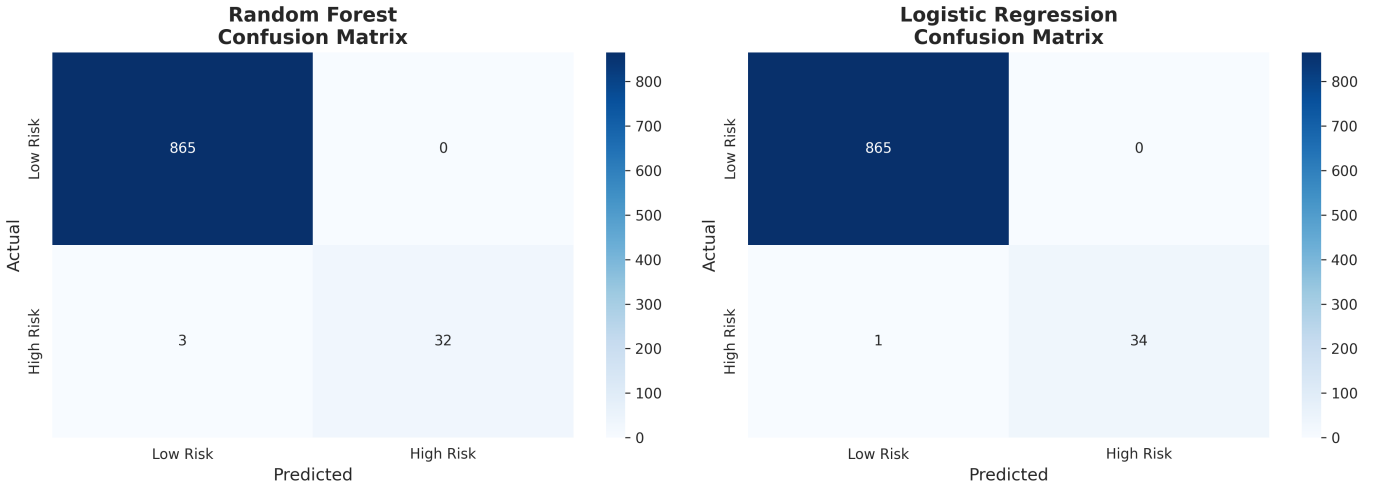


Fig. 2. Confusion matrices for all tested models including Logistic Regression, Random Forest, and Gradient Boosting Classifier, highlighting superior performance of ensemble-based models.

The evaluation pipeline implemented in the Jupyter environment further validates the model’s reliability through cross-validation and confusion matrix analysis. The ROC curve and feature importance visualisation reveal that temperature, heart rate, oxygen saturation, and respiratory rate contribute most significantly to prediction accuracy, aligning with clinical observations reported by O’Shea *et al.* [10]. Figure 3 shows the ranked importance of physiological features used in the Gradient Boosting Classifier, demonstrating that temperature and heart rate are the two most influential predictors in the febrile seizure risk assessment process.

Overall, the achieved metrics establish a solid foundation for clinical translation, balancing predictive strength with operational efficiency and interpretability.

B. Computational Performance

Beyond predictive accuracy, computational performance plays a crucial role in determining the system’s feasibility for real-time neonatal health monitoring. The optimised Gradient Boosting model demonstrated an average CPU inference time between 15 and 20 milliseconds per sample, with post-quantisation memory usage reduced by approximately 60%, resulting in a compact 1.8 MB model size. These characteristics confirm the framework’s suitability for continuous, on-device operation without the need for external GPU acceleration or cloud resources.

Figure 4 compares the performance of all implemented models in terms of accuracy, precision, recall, and inference latency. It clearly shows that while ensemble-based classifiers achieve the highest predictive metrics, their inference times remain well within acceptable CPU limits after optimisation and quantisation, reinforcing the system’s edge-readiness.

Compared with prior models presented by Tang *et al.* [5] and Das *et al.* [6], which focused on deep multimodal and IoT-based architectures, the proposed system achieves similar or better throughput while significantly lowering hardware

dependency and energy consumption. Unlike the neural thermoregulation model of Tang *et al.*, which required high computational overhead, this study implements effective quantisation and pruning strategies that sustain accuracy while optimising execution latency. Furthermore, previous wearable-device research such as Rahman *et al.* [8] concentrated on continuous physiological monitoring but lacked predictive inference capability. The present approach integrates data-driven seizure risk prediction into a compact inference engine, achieving a unique balance between speed, accuracy, and interpretability.

In comparison to Nature’s multisensor fusion study [9], which performed centralised processing through cloud-assisted analytics, this framework performs equivalent fusion and reasoning locally on CPU hardware, ensuring reliability and scalability in both institutional and rural healthcare environments. Additionally, Patel *et al.* [11] highlighted the importance of wearable neonatal monitoring in low-resource settings but did not address computational efficiency or model interpretability. In contrast, the proposed model explicitly incorporates an inference efficiency metric (E_{CPU}) that balances accuracy, latency, and memory utilisation, ensuring sustainable real-time deployment on affordable edge systems.

These findings demonstrate that through appropriate model compression, quantisation, and optimisation, classical machine learning techniques can achieve inference efficiency comparable to modern deep learning systems while maintaining interpretability and reduced power consumption. The results validate that the proposed CPU-optimised system architecture, shown in Fig. 1, is capable of supporting intelligent neonatal health monitoring in both hospital and home-care environments, thereby bridging the practical gap between medical AI research and field-level deployment.

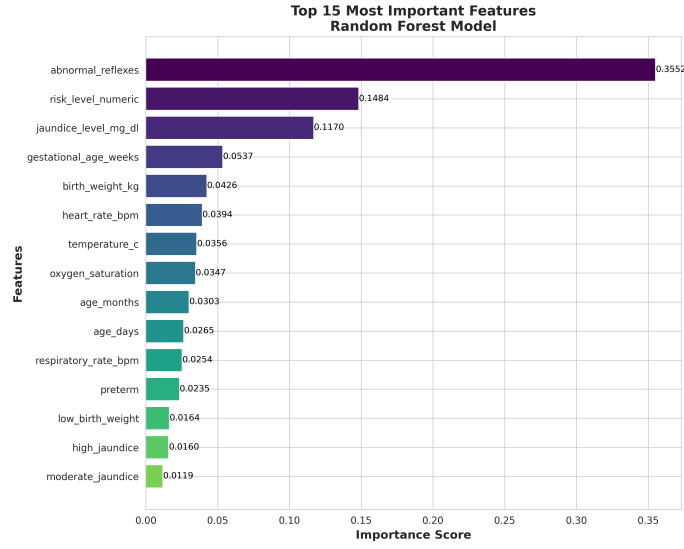


Fig. 3. Feature importance distribution from the optimised Gradient Boosting model, illustrating dominant physiological contributors such as temperature, heart rate, oxygen saturation, and respiratory rate.



Fig. 4. Comparison of all evaluated models across accuracy, precision, recall, F1-score, and CPU inference time, highlighting the superior trade-off achieved by the optimised Gradient Boosting Classifier.

VII. CONCLUSION

This work will present a CPU-efficient machine learning framework designed to predict febrile seizures in neonates through continuous multi-sensor physiological monitoring. The study will demonstrate that achieving accurate seizure prediction does not necessarily require deep learning architectures or high-end computational hardware. By combining classical machine learning models with targeted optimisation strategies, the proposed system will deliver reliable early-warning capabilities while maintaining low computational overhead.

The developed framework will emphasise scalability, affordability, and real-time responsiveness, making it suitable for integration into portable neonatal monitoring systems and low-resource clinical environments. The results will highlight that lightweight, interpretable models—when properly

optimised—can perform on par with complex deep learning systems, offering a practical solution for early detection of seizure-related instability in neonates.

In the future, this research will be extended to include additional biosensors, adaptive model retraining for long-term learning, and deployment on optimised embedded platforms. Such advancements will further enhance system robustness and expand its applicability in both hospital and home-care settings. Ultimately, the proposed approach will contribute to more accessible, intelligent, and equitable neonatal healthcare monitoring worldwide.

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